HOSTING REQUEST · OCTOBER 2026

SLIMS — server requirements

Street Lighting Store, Issuance & Installation Management System: a web portal and API, a SQL Server database, a folder of geo-tagged installation photographs, and the SLIMS Field Android app used by Lighting Officers at the pole. **Two virtual machines are requested.**

Prepared for
Municipal Town Admin Office
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1 · Web / application server — PUBLIC IP

CORES	RAM	STORAGE	LIVE IP
8 vCPU	16 GB	500 GB SSD, expandable	2 public, HTTPS only

OS: Windows Server 2022 Standard ×64 (2019 acceptable). The 500 GB covers the operating system and the installation photographs, and must be expandable later without rebuilding the VM. Two public addresses are requested: one for the service, one held for failover or a second site.

2 · Database server — PRIVATE ONLY

CORES	RAM	STORAGE	LIVE IP
12 vCPU	16 GB	200 GB SSD, expandable	0 private network

OS: Windows Server 2022 Standard ×64 (2019 acceptable). **SQL Server Standard is requested** — the free Express edition uses only about 1.4 GB of memory and four cores however large the VM is, so this specification pays off only on Standard. The 200 GB holds the database, its transaction log and local backups for many years — the database grows by roughly 1 GB per 100,000 items.

What fills the web server's 500 GB

Each installation stores a BEFORE and an AFTER photograph (1920×1080 JPEG) plus the server's stamped copy of each — about 2.5 MB per installation. Nothing else in the system grows.

10,000	25 GB
50,000	125 GB
100,000	250 GB

Installations recorded, against photograph storage used. At this rate **500 GB holds about 200,000 installations**, and the disk can be grown later. The database stays small — roughly 1 GB per 100,000 items — because photographs are files on disk, not rows in the database.

Bandwidth

MEASURE	FIGURE	WHERE IT COMES FROM
Sustained	20–30 Mbps	20 officers uploading at once ≈ 20 Mbps inbound
Burst	50 Mbps	Shift start, when queued captures sync
Monthly transfer	20–25 GB	≈ 7 GB per 5,000 installations, plus portal traffic
Per installation	≈ 1.4 MB	Two photographs, sent from the phone

Software required on the servers

Everything else ships inside our deployment package.

SERVER	NEEDS
Web / application	Node.js 20 LTS or newer · IIS with URL Rewrite + ARR, or Nginx, to terminate HTTPS · antivirus exclusion on the uploads folder. **No .NET runtime** — the API is self-contained.
Database	SQL Server 2019/2022 (Standard requested, Express workable) · SSMS · mixed-mode authentication · antivirus exclusion on SQL data and log folders.
Both	**NTP time synchronisation enabled.** Photographs are checked against server time; a drifting clock makes the app refuse captures.

Ports

Portal (3001) and API (5091) stay on 127.0.0.1 behind the proxy and are never exposed.

PATH	PORT	PURPOSE
Internet → web	443, 80	Users and phones; 80 redirects to 443
Web → database	1433	Private network only
Admin → both	3389	RDP, restricted addresses
Web → outbound	587 / 25, 443	Alert e-mail; SMS gateway and updates

Also needed from PITB

- **A web address and its security certificate** — for example streetlight.pera.gop.pk, with an SSL certificate that PITB keeps renewed. The officers' app is downloaded from SLIMS itself, not from an app store, so this address is how phones install and update it; the browser also refuses to give an officer's location on an unsecured address.
- **Backups** — the database and the photographs copied every night and kept for 30 days, plus a full copy of both servers once a week. One restore is tested every three months, so we know the backups work.
- **Three kinds of login** — one administrator login per server for installing and updating, one account the application runs under, and one database login that can reach only the SLIMS database.

Why the public addresses are needed

Lighting Officers work in the field, so the web server has to be reachable over the internet: two public addresses, a DNS record and an SSL certificate. No VPN is required on the phones — the Android app connects straight to the published address over HTTPS.

Figures measured on the running system: photograph size, upload payload and database growth.